BrightLearn SQL Aggregate Functions & Operators

Name : Amanda Zisanda

Surname : Ntisana

Student No : 280 500 1778

Course : Data Analytics and Data Sciences

Module : BrightLearn Basic SQL 2

Due Date : 01 May 2026.

01 THE STUDENTS TABLE

TABLE students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

01. List all distinct departments in the students table.

```
SELECT DISTINCT department
FROM students
```

Output:

department
IT
HR
Finance

02. Get the average age of students per department.

```
SELECT department, AVG (age) AS avg_age
FROM students
GROUP BY department;
```

Output:

department	avg_age
IT	20.5
HR	22
Finance	23

3. Show departments with more than 1 student.

```
SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;
```

Output

department	student_count
IT	2
HR	2

4. Get all students whose age is between 21 and 23.

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 and 23.
```

Output

~~department~~

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21.

```
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

Output

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

02. THE COURSES TABLE

TABLE courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6. Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department, SUM (credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;
```

Output

department	total_credits
IT	11

7. List all courses that do not have 4 credits.

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE NOT credits = 4;
```

Output

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Show the top 3 courses by credits in descending order.

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

Output

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

03 THE ENROLLMENTS TABLE

TABLE enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9. Get the maximum, minimum, and average grade across all enrollments.

```
SELECT
  MAX(grade) AS max_grade,
  MIN (grade) AS min_grade,
  AVG (grade) AS avg_grade
FROM enrollments;
```

Output

max_grade	min_grade	avg_grade
90	78	84,6

10. Count how many enrollments exist per course.

```
SELECT course_id, COUNT(*) AS enrollment_count
FROM courses
GROUP BY course_id;
```

Output

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 THE SALARIES TABLE

TABLE salaries

employer_id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11. Find total salary and total bonus per department

```
SELECT department, sum salary AS total_salary,
sum bonus AS total_bonus
FROM salaries
GROUP BY department;
```

output:

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. Show departments where average salary is above 55,000.

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

Output

department	avg_salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 60 000.

```
SELECT employee_id, name, salary, bonus, (salary+bonus)
AS total_compensation
FROM salaries
WHERE (salary+bonus) > 60 000;
```

Output

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	4 000	74 000
4	Tyke	62 000	5500	67 500

05 THE PROJECTS TABLE

TABLE projects

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

14. Show total and average budget per department. Only include departments with average budget above 70 000.

```
SELECT department,
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70 000;
```

Output

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50 000 and 120 000, excluding the Marketing department.

```
SELECT   project_id, project_name, department, budget
FROM     projects
WHERE    budget BETWEEN 50 000 and 120 000
AND NOT  department = 'Marketing';
```

Output

project_id	project_name	department	budget
~~AI App~~ 1	AI App	IT	120 000
~~Pro~~ 2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000